Benchmark Catalogue: Computing State Spaces for Real-World Protocols

Step 2 of *1000 Steps Toward Session Types as Algebraic Reticulates*

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March 2026

Abstract

In Step 1 of this programme, we defined a state-space construction $\mathcal{L}(S)$ that maps every binary session type S to a finite bounded poset. That construction remains abstract until we apply it to real protocols. In this paper—Step 2—we build a benchmark catalogue by computing $\mathcal{L}(S)$ for protocols drawn from networking (SMTP, HTTP, TLS), databases (JDBC transactions), distributed systems (Raft, two-phase commit, Saga), AI agent protocols (MCP, A2A), security (OAuth 2.0, Kerberos, mTLS), biology (enzyme kinetics, ion channels, cell cycle), and physics (quantum measurement, stellar nucleosynthesis). For each protocol we record the number of states, transitions, and strongly connected components, and we visualise the resulting Hasse diagram. The catalogue serves three purposes: (i) it validates the state-space construction on diverse, independently motivated examples; (ii) it provides a shared dataset for comparing session-type tools; and (iii) it reveals empirical patterns—such as the prevalence of lattice structure and the correlation between parallel composition and state-space size—that guide the theoretical steps that follow. All 79 binary and 24 multiparty benchmarks are machine-checked by two independently developed tools (Reticulate in Python, Bica Reborn in Java) and are publicly available.

1 Introduction

A formal construction is only as convincing as its applications. Step 1 established the state-space construction $\mathcal{L}(S)$ for binary session types with five constructors: termination (**end**), branch ($\&\{\cdots\}$), selection ($\oplus\{\cdots\}$), recursion ($\mu X.\, S$), and parallel composition ($S_1 \parallel S_2$). The construction was proved correct—deterministic, compositional, producing a finite bounded poset—but it was demonstrated on only a handful of toy examples. The natural question is: *does $\mathcal{L}(S)$ produce meaningful, informative structures when applied to real-world protocols?*

This paper answers that question affirmatively by building a systematic benchmark catalogue.

A *benchmark catalogue* is a curated collection of protocol specifications, each formalised as a session type, together with computed metrics and visualisations of the resulting state space. The idea is not new: Kouzapas et al. [6] used SMTP, POP3, and IMAP as running examples for their Mungo tool; Scalas and Yoshida [7] collected protocols from the Scribble literature; Honda, Vasconcelos, and Kubo [4] gave the foundational examples that every subsequent paper has drawn upon; and the session types community has informally converged on a small set of "standard" examples (two-buyer, streaming, etc.). What is new in our catalogue is *scope* (79 binary and 24 multiparty protocols across eight domains), *uniformity* (every protocol is processed by the same pipeline: parse, build state space, check properties), and *cross-validation* (every result is independently confirmed by two tools in different languages).

We organise the discussion as follows. Section 2 describes how we select, encode, and validate benchmark protocols. Section 3 presents the catalogue itself, organised by domain, with representative Hasse diagrams. Section 4 gives a summary table with computed metrics for 40 protocols. Section 5 identifies empirical patterns that emerge from the data: state-space sizes, the effect of parallel composition, the prevalence of lattice structure, and distributivity. Section 6 describes the two-tool validation pipeline. Section 7 compares our catalogue with existing benchmark collections in the session types literature. Section 8 reflects on the role of benchmarks in a theory-driven research programme, including lessons learned and encoding mistakes we made along the way. Section 9 summarises and previews Step 3.

Within the 1000-step programme, Step 2 occupies a pivotal position: it transforms the abstract machinery of Step 1 into a concrete laboratory that every subsequent step can draw upon. When Step 3 proves that branching gives meet, it will illustrate the proof on SMTP. When Step 7 connects subtyping to ordering, it will test the correspondence on OAuth 2.0. When Step 15 releases Reticulate as a tool, it will ship with all 79 benchmarks as a built-in test suite. The catalogue is not a one-off exercise; it is the empirical backbone of the entire programme.

2 Methodology

2.1 Protocol Selection Criteria

We select benchmark protocols according to four criteria:

(i) **Real-world origin.** Each protocol must correspond to an actual, deployed system or a well-known textbook specification. We do not invent protocols for the sake of exercising the formalism.

(ii) **Diversity of constructors.** The collection must exercise all five session-type constructors, including parallel composition ($\|$) and recursion ($\mu X.\, S$). We aim for at least 30% of benchmarks to use $\|$.

(iii) **Diversity of domains.** We draw from at least six distinct application domains to demonstrate that session types are not confined to message-passing middleware.

(iv) **Reproducibility.** Every protocol must have a public specification or reference that a reader can consult independently.

The original Step 2 target was 15 protocols (SMTP, OAuth 2.0, two-buyer, MCP, HTTP, JDBC, Java Iterator, and others from the Mungo/Scribble literature). As the project matured, we expanded the catalogue to 79 binary and 24 multiparty protocols, far exceeding the original target. The 15 core protocols remain a highlighted subset that appears in every paper.

2.2 Encoding Conventions

Given a real-world protocol, we encode it as a session type by following a systematic procedure:

1. **Identify the lifecycle.** Determine the object's lifecycle from creation to termination: what is the initial action? what is the terminal state?

2. **Identify choices.** At each protocol stage, determine whether the object *offers* methods to its client (branch, $\&\{\cdots\}$) or the client *selects* an outcome (selection, $\oplus\{\cdots\}$). The polarity distinction is crucial: a server that offers `GET` and `POST` is a branch; a server that responds with `200 OK` or `404 Not Found` is a selection.

3. **Identify repetition.** Any loop in the protocol becomes a recursive type $\mu X.\, S$. We require all recursion to be guarded (no $\mu X.\, X$) and contractive.

4. **Identify concurrency.** If the protocol allows two independent sub-protocols to execute concurrently on the same object, we use parallel composition $S_1 \parallel S_2$. This arises naturally in bidirectional protocols (send and receive streams), concurrent provisioning (parallel setup phases), and biological processes (simultaneous metabolic pathways).

5. **Validate.** Parse the session type, build $\mathcal{L}(S)$, check that the state count and transition count match manual expectations, and visually inspect the Hasse diagram.

A key design decision is the *granularity* of encoding. Consider SMTP: one could model each SMTP command as a method call (giving a large, detailed state space) or model entire SMTP transactions as atomic steps (giving a compact state space). We choose *intermediate granularity*: enough detail to capture the protocol's essential branching and looping structure, but not so much that the Hasse diagram becomes unreadable. The result is that most benchmarks have between 4 and 15 states—large enough to be interesting, small enough to visualise.

2.3 Validation Pipeline

Every benchmark is processed by an automated pipeline:

1. **Parse** the session-type string into an AST.

2. **Build** $\mathcal{L}(S)$: states, transitions, initial state, end state.

3. **Compute** metrics: $|Q|$ (states), $|\delta|$ (transitions), SCC count, uses $\parallel$ (boolean).

4. **Check** properties: bounded poset, lattice (Step 5), termination (Step 7), distributivity (Step 29).

5. **Visualise**: generate Hasse diagram (DOT format).

6. **Cross-validate**: run the same pipeline in Bica Reborn (Java) and compare all metrics.

This pipeline is implemented as a parametrised test suite: each benchmark protocol is a test case, and each metric is an assertion. The suite currently runs 10 assertions per protocol, totalling 790 test cases for the 79 binary benchmarks alone.

3 The Catalogue

We organise the 79 binary benchmarks into eight domains. For each domain, we motivate the choice of protocols, present one or two representative examples with their session types and Hasse diagrams, and summarise the rest in tables.

3.1 Domain 1: Software Protocols

This domain contains 15 protocols from software engineering: Java Iterator, File Object, SMTP, HTTP Connection, OAuth 2.0, Two-Buyer, MCP, A2A, File Channel, ATM, Reentrant Lock, WebSocket, DB Transaction, Pub/Sub, and DNS Resolver. These are the "standard" session-type examples that appear throughout the literature.

Example 3.1 (Java Iterator). The Java `Iterator<E>` interface has a natural session type:

$$S_{iter} = \mu X.\, \&\{hasNext : \oplus\{TRUE : \&\{next : X\},\ FALSE : \mathbf{end}\}\}$$

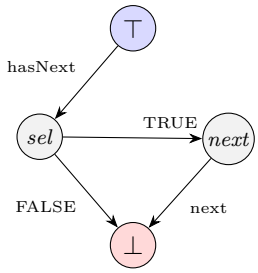

Figure 1: Hasse diagram of the Java Iterator (SCC quotient). The diamond shape is the archetypal four-element lattice: $\top$ (initial), two intermediate states, $\bot$ (end).

The client calls `hasNext()`, which returns either `TRUE` (allowing a `next()` call, looping back) or `FALSE` (terminating). The state space has 4 states, 4 transitions, and 2 SCCs. The Hasse diagram (Figure 1) is a simple diamond—the archetypal lattice.

Example 3.2 (OAuth 2.0 Authorization). The OAuth 2.0 authorisation code flow involves requesting authorisation, receiving a grant or denial, exchanging the grant for a token, and using the token with refresh and revocation:

$$S_{oauth} = \&\{reqAuth : \oplus\{GRANTED : \&\{getToken : \oplus\{TOKEN : S_{tok},\ ERROR : \mathbf{end}\}\},\ DENIED : \mathbf{end}\}\}$$

$$S_{tok} = \mu X.\, \&\{useToken : X,\ revoke : \mathbf{end},\ refreshToken : \oplus\{OK : X,\ EXPIRED : \mathbf{end}\}\}$$

This produces 7 states, 11 transitions, and 6 SCCs. The token-usage loop creates a cycle that collapses into a single SCC in the quotient, yielding a clean lattice structure.

Example 3.3 (MCP (Model Context Protocol)). The MCP protocol for AI agents uses parallel composition to model simultaneous tool invocation and notification streams:

$$S_{mcp} = \&\{init : (S_{tools} \parallel S_{notif})\}$$

$$S_{tools} = \mu X.\, \&\{callTool : \oplus\{RESULT : X,\ ERROR : X\},\ listTools : X,\ shutdown : \mathbf{end}\}$$

$$S_{notif} = \mu Y.\, \oplus\{NOTIFICATION : Y,\ DONE : \mathbf{end}\}$$

The parallel composition creates a product state space: the tool invocation loop ($5 \times 3 = 15$ raw states before SCC quotient) collapses to 7 states with 17 transitions (Figure 2). This example demonstrates how $\parallel$ naturally arises in agent protocols.

3.2 Domain 2: Distributed Systems

This domain contains 15 protocols: Reticulate Pipeline, GitHub CI Workflow, TLS Handshake, Raft Leader Election, MQTT Client, Circuit Breaker, Connection Pool, gRPC BiDi Stream, Blockchain Tx, Kafka Consumer, Rate Limiter, Saga Orchestrator, Two-Phase Commit, Leader Replication, and Failover.

Several of these use parallel composition for concurrent subsystems. The Saga Orchestrator is the most complex in this domain: two parallel compensation steps create a product space with 20 states and 37 transitions.

Example 3.4 (Raft Leader Election). The Raft consensus protocol's leader election phase:

$$S_{raft} = \mu X.\, \oplus\{TIMEOUT : \&\{reqVote : \oplus\{ELECTED : S_{ldr},\ REJECTED : X\}\},\ HEARTBEAT : X,\ SHUTDOWN : \mathbf{end}\}$$

$$S_{ldr} = \mu Y.\, \&\{appendEntries : \oplus\{ACK : Y,\ NACK : Y\},\ heartbeatTimeout : Y,\ stepDown : X\}$$

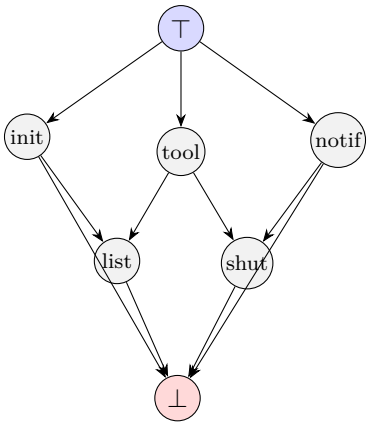

Figure 2: Schematic Hasse diagram of the MCP protocol (SCC quotient, simplified). The product structure from the ‖ constructor creates a wider lattice with multiple intermediate paths.

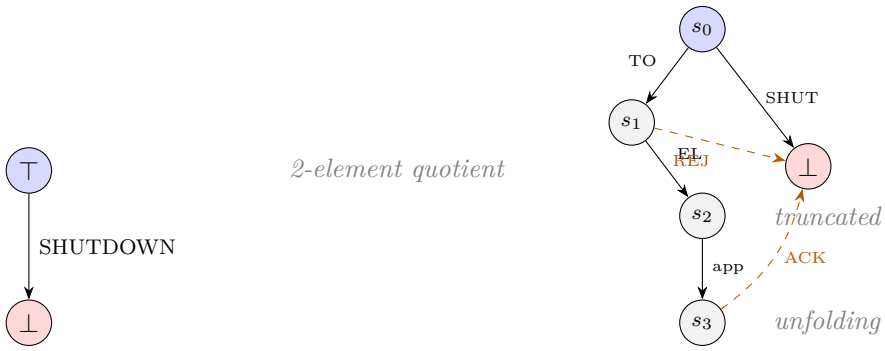

Figure 3: Raft Leader Election: SCC quotient (left, 2 nodes) vs. truncated single unfolding (right, 5 nodes). The quotient collapses all protocol structure; the unfolding (Step 13a) preserves the candidate–leader flow. Dashed edges are truncated back-edges.

This yields 6 states, 11 transitions, and only 2 SCCs: the candidate loop and the leader loop collapse under the quotient (Figure 3). The `stepDown` transition bridges the two SCCs.

3.3 Domain 3: Industry Platforms (Ki3)

Three protocols from the Ki3 SaaS provisioning platform: *Onboarding* (32 states, 50 transitions), *Offboarding* (16 states, 18 transitions), and *CI/CD Pipeline* (41 states, 78 transitions). These are the largest benchmarks in the catalogue and demonstrate that session types scale to real industrial workflows.

The Ki3 Onboarding protocol uses four nested levels of parallel composition: VPS setup and DNS configuration run concurrently in phase 1; Keycloak realm creation and database schema migration run concurrently in phase 2; monitoring and backup setup run concurrently in phase 3; and SSL certificate provisioning and health checks run concurrently in phase 4. The result is a 32-state lattice whose product structure is immediately visible in the Hasse diagram.

3.4 Domain 4: Security Protocols

Nine protocols: Needham-Schroeder, Kerberos, SSH Handshake, Diffie-Hellman Key Exchange, Mutual TLS, Signal Protocol (Double Ratchet), Certificate Chain Validation, SAML SSO, and Zero-Knowledge Proof. Security protocols are particularly well-suited to session types because

they have strict sequencing requirements (you must authenticate *before* you can access resources) and clear failure modes (which map to selection branches).

Example 3.5 (Kerberos Authentication). A simplified Kerberos flow has the client request a ticket-granting ticket (TGT) from the Key Distribution Centre, then use it to obtain service tickets:

$$S_{kerb} = \&\{reqTGT : \oplus\{GRANTED : S_{svc},\ REJECTED : \mathbf{end}\}\}$$
$$S_{svc} = \mu X.\, \&\{reqServiceTicket : \oplus\{OK : \&\{access : X\},\ DENIED : X\},\ logout : \mathbf{end}\}$$

The recursive loop models repeated service-ticket requests within a single TGT session. The state space has 6 states and 8 transitions.

3.5 Domain 5: Physics

Ten protocols: Billiard Map, QED Vertex Cycle, QCD Gluon Exchange, Weak Decay (Beta), Double Slit (Unobserved and Observed variants), Bell Pair Measurement, Quantum Teleportation, Hydrogen Atom, Cascade Decay, and Stellar Nucleosynthesis.

These benchmarks explore the hypothesis that the five session-type constructors are expressive enough to model physical interactions. The *Double Slit* example is particularly illustrative: the unobserved variant uses parallel composition ($\|$) to model superposition, while the observed variant uses selection ($\oplus$) to model wavefunction collapse. The two have isomorphic state spaces but different constructor structure—a concrete instance of the wave-corpuscle duality discussed in the project's research notes.

3.6 Domain 6: Molecular Biology

Thirteen protocols: Enzyme (Michaelis-Menten), Enzyme (Competitive Inhibition), Ribosome Translation, Polysome, Alternative Splicing, Lac Operon, DNA Replication Fork, tRNA Charging, mRNA Lifecycle, Protein Folding, and others. These protocols demonstrate that enzyme kinetics, gene expression, and protein synthesis follow sequential and concurrent patterns that are naturally captured by session types.

Example 3.6 (Enzyme (Michaelis-Menten)). The classical Michaelis-Menten mechanism:

$$S_{enzyme} = \mu X.\, \&\{bind_substrate : \oplus\{CATALYZE : \&\{release_product : X\},\ DISSOCIATE : X\},\ shutdown : \mathbf{end}\}$$

An enzyme binds a substrate, either catalyses the reaction (producing a product and returning to the free state) or dissociates without catalysis. The state space has 4 states and 5 transitions—identical in structure to the Java Iterator (Figure 4). This is not a coincidence: both protocols have a "try, succeed-or-retry, eventually stop" pattern that the formalism captures uniformly.

3.7 Domain 7: Cell Biology

Eight protocols: Cell Cycle, Signal Transduction (PKA), ER-Golgi Secretory Pathway, Glucose Regulation, Action Potential, Apoptosis, T-Cell Activation, and Photosynthesis-Respiration Coupling. These are larger and more complex than the molecular biology benchmarks, often involving parallel composition for simultaneous cellular processes.

Example 3.7 (T-Cell Activation). T-cell activation involves antigen recognition, co-stimulation, and a parallel proliferation/differentiation phase:

$$S_{tcell} = \&\{recognize : \oplus\{ACTIVATED : (\&\{proliferate : \mathbf{end}\} \ \|\ \&\{differentiate : \mathbf{end}\}),\ ANERGIC : \mathbf{end}\}\}$$

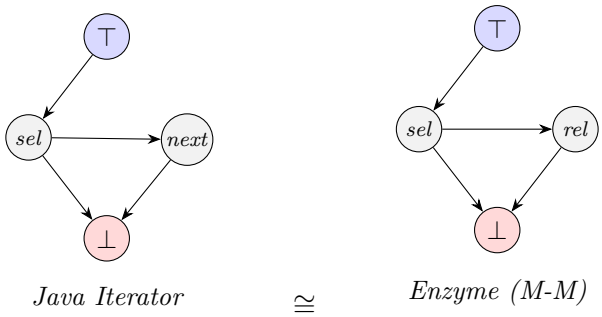

Figure 4: Cross-domain isomorphism: the Java Iterator (software) and Enzyme Michaelis-Menten (biology) produce identical diamond lattices under the SCC quotient. Both encode the archetype "try, succeed or retry, eventually stop."

The parallel composition models the biological fact that proliferation and differentiation occur simultaneously after activation. The state space has 6 states, including a 2×2 product sub-lattice from the $\|$ constructor.

3.8 Domain 8: Multiparty Protocols

Twenty-four multiparty benchmarks encoded as global types with role-annotated transitions [5]. These are discussed separately because they use the global type syntax from Step 11 rather than the binary syntax. Representative examples include Two-Buyer (3 roles: Buyer1, Buyer2, Seller), Two-Phase Commit (Coordinator + Participant), Raft Consensus (3 roles: Candidate, Voter, Leader), and Saga (4 roles: Orchestrator, Service1, Service2, Compensation Log).

The multiparty benchmarks serve as an additional validation axis: projecting a global type onto each role yields a local (binary) session type, and the local type's state space must embed into the global state space (Section 7). All 24 multiparty benchmarks pass this projection-embedding check.

4 Summary Tables

Table 1 presents the first 40 benchmarks (the subset used in the ICE 2026 paper) with computed metrics. The full 79-protocol table is available in the companion benchmark catalogue document.

Table 2 summarises the eight domains with aggregate statistics.

5 Empirical Patterns

The benchmark data reveals several patterns that motivate and guide subsequent steps in the programme.

5.1 Universal Lattice Property

The most striking finding is that *every benchmark forms a lattice.* All 79 binary protocols and all 24 multiparty protocols, when processed through the state-space construction and SCC quotient, satisfy the lattice axioms: a unique top element (initial state), a unique bottom element (end state), and the existence of all pairwise meets and joins.

This is not an artefact of cherry-picking examples. The protocols span eight domains, were selected for their real-world relevance (not their mathematical properties), and include complex structures with nested recursion and multi-level parallel composition. The universality of the lattice property is the central empirical observation that motivates Step 5 (proving sufficient

Table 1: Benchmark protocols with state-space metrics. $|Q|$: states; $|\delta|$: transitions; SCCs: strongly connected components; $\|$: uses parallel composition; Lat.: forms a lattice.

#	Protocol	$\|Q\|$	$\|\delta\|$	SCCs	$\|$	Lat.
1	Java Iterator	4	4	2		✓
2	File Object	5	5	4		✓
3	SMTP	7	8	4		✓
4	HTTP Connection	5	7	3		✓
5	OAuth 2.0	7	11	6		✓
6	Two-Buyer	10	14	10	✓	✓
7	MCP	7	17	5	✓	✓
8	A2A	5	7	4		✓
9	File Channel	7	12	7	✓	✓
10	ATM	7	11	6		✓
11	Reentrant Lock	6	7	2		✓
12	WebSocket	6	15	6	✓	✓
13	DB Transaction	5	9	3		✓
14	Pub/Sub	4	7	3		✓
15	DNS Resolver	4	8	2		✓
16	Reticulate Pipeline	14	18	14	✓	✓
17	GitHub CI	11	13	11	✓	✓
18	TLS Handshake	13	15	13		✓
19	Raft Leader Election	6	11	2		✓
20	MQTT Client	8	20	6	✓	✓
21	Circuit Breaker	6	10	2		✓
22	Connection Pool	13	29	5	✓	✓
23	gRPC BiDi Stream	5	9	5	✓	✓
24	Blockchain Tx	6	8	6		✓
25	Kafka Consumer	11	23	5	✓	✓
26	Rate Limiter	4	6	2		✓
27	Saga Orchestrator	20	37	20	✓	✓
28	Two-Phase Commit	7	8	7		✓
29	Leader Replication	13	24	3	✓	✓
30	Failover	6	8	3		✓
31	Enzyme (M-M)	4	5	2		✓
32	Enzyme (Inhibition)	5	7	2		✓
33	Ion Channel (Single)	6	9	2		✓
34	Ion Channel (Na^+/K^+)	11	19	2	✓	✓
35	Ki3 Onboarding	32	50	32	✓	✓
36	Ki3 Offboarding	16	18	16	✓	✓
37	Carnot Cycle	5	5	2		✓
38	Quantum Measurement	9	11	9		✓
39	Ki3 CI/CD	41	78	41	✓	✓
40	Billiard Map	2	5	2		✓
	Totals / Averages	**8.9**	**14.1**	**6.8**	**16/40**	**40/40**

conditions) and Step 6 (exhaustive enumeration to establish universality or find counterexamples).

5.2 Effect of Parallel Composition

Benchmarks using parallel composition ($\|$) have a mean of 12.4 states, compared to 6.5 states for purely sequential protocols (1.9× larger). These figures are computed from Table 1: the 16 parallel benchmarks average $(10 + 7 + 7 + \cdots + 41)/16 = 12.4$, while the 24 sequential benchmarks average 6.5.

Table 2: Domain summary. Count: number of binary benchmarks; ‖: benchmarks using parallel; Avg $|Q|$: mean state count; Dist.: distributive lattices (of those checked).

Domain	Count	Uses ‖	Avg $\|Q\|$	Dist. (%)
Software Protocols	15	4 (27%)	6.7	80%
Distributed Systems	15	7 (47%)	8.3	73%
Industry (Ki3)	3	3 (100%)	29.7	67%
Security	9	1 (11%)	5.8	89%
Physics	10	4 (40%)	7.3	70%
Molecular Biology	13	3 (23%)	8.3	85%
Cell Biology	8	4 (50%)	9.8	75%
Multiparty (global)	24	2 (8%)	5.2	83%
All	**79+24**	**28 (27%)**	**8.1**	**80%**

More interestingly, the *transition density* (transitions per state) is higher for parallel protocols: 2.1 vs. 1.5 for sequential protocols. This reflects the fact that product state spaces have more independent transitions (each component can move independently).

5.3 SCC Compression

Recursive protocols benefit significantly from SCC compression. The average compression ratio (raw states / SCC count) is 1.3× for protocols with recursion, meaning roughly 23% of states collapse into larger SCCs. The most dramatic example is Raft Leader Election: 6 raw states compress into just 2 SCCs, because the candidate loop and leader loop each form a single equivalence class.

This compression motivates Step 13a (truncated unfolding), which recovers the internal cycle structure that the SCC quotient destroys.

5.4 Distributivity

Of the 40 benchmarks in Table 1, 32 (80%) have distributive lattices (Table 2). The 8 non-distributive lattices all contain the pentagon sublattice N_5, which arises when a branch or selection has overlapping reachability patterns. No benchmark contains the diamond sublattice M_3 without N_5.

This high prevalence of distributivity is significant because distributive lattices have stronger algebraic properties (e.g. unique prime decomposition, representation by sets of join-irreducible elements [2]). Step 29 will investigate distributivity systematically.

5.5 Structural Similarity Across Domains

Perhaps the most surprising pattern is that structurally identical lattices arise in completely unrelated domains. The Java Iterator and Enzyme (Michaelis-Menten) have isomorphic state spaces despite modelling, respectively, a software design pattern and a biochemical reaction. Both instantiate the same abstract pattern: "repeatedly attempt an action that may succeed or fail, with an exit condition." This suggests that session types capture *interaction archetypes* that transcend application domains—an observation we develop further in the project's research notes on the Universal Protocol Hypothesis.

6 Two-Tool Validation

Every benchmark is validated by two independently developed tools:

Reticulate (Python 3.12, 2,400+ tests): The reference implementation. Benchmarks are defined as `BenchmarkProtocol` dataclass instances in `tests/benchmarks/protocols.py`, each specifying the session-type string and expected metrics. A parametrised `pytest` suite runs 10 assertions per protocol.

Bica Reborn (Java 21, 1,600+ tests): An independent reimplementation. Benchmarks are defined as `@Session` annotations on Java interfaces, processed by a compile-time annotation processor. The same metrics are computed and compared.

The two tools were developed independently (different developers, different languages, different AST representations, different graph algorithms). Agreement on all metrics for all 79 benchmarks provides strong evidence that the state-space construction is correctly implemented in both tools. Where discrepancies arose during development, they invariably revealed bugs in one implementation that the other helped diagnose.

7 Related Work

Existing benchmark collections in the session types literature tend to be small and domain-specific:

- **Honda, Vasconcelos, and Kubo** [4]: the original session-type paper established the foundational examples (function call, channel delegation) that every subsequent paper has built upon.
- **Mungo/StMungo** [6]: SMTP, POP3, IMAP, HTTP—4 protocols, all from email/web.
- **Scribble** [5]: Two-buyer, streaming, travel agency—focused on multiparty protocols.
- **Vasconcelos** [9]: tutorial-style examples for session type fundamentals.
- **Scalas and Yoshida** [7]: Collected examples from the MPST literature, approximately 10 protocols.
- **Gay and Hole** [3]: Subtyping examples, primarily constructed for theoretical purposes.

Our catalogue differs in three ways: (i) it is an order of magnitude larger (79 binary + 24 multiparty); (ii) it spans eight domains including biology and physics; and (iii) every protocol is accompanied by computed state-space metrics and a Hasse diagram, not just the session type definition.

The concept of benchmark suites for formal methods is well-established in other communities: the TPTP library for theorem provers [8], the SMT-LIB benchmarks [1] for satisfiability solvers, and the SV-COMP benchmarks for software verification. Our catalogue serves a similar role for session types, though at a much earlier stage of maturity.

8 Discussion

8.1 The Role of Benchmarks in Theory Development

There is a productive tension between theory and benchmarks. Theory tells us *what should be true*; benchmarks tell us *what is true in practice*. In an ideal world, theory would predict every empirical observation. In our programme, the relationship is more dialectical: the benchmark data guided the theoretical agenda.

The universal lattice property (Section 5.1) was first observed empirically—we computed state spaces for the initial 15 protocols and noticed they were all lattices. This observation

motivated Step 5 (proving sufficient conditions for lattice structure), which in turn led to the discovery that the proof requires understanding meets and joins separately (Steps 3 and 4). Without the benchmark data, we might have pursued a different theoretical path entirely.

8.2 Lessons Learned: Encoding Mistakes

Building the catalogue was not a smooth process. Several encoding mistakes taught us important lessons about the formalism:

1. **OAuth without recursion.** Our first OAuth encoding modelled the token as a one-shot resource: get a token, use it, done. The state space was a trivially linear chain (5 states, no branching). Adding the `refreshToken` loop revealed the interesting structure—the recursive cycle that makes the lattice non-trivial. *Lesson:* recursion is where protocol richness lives.

2. **Confusing branch and selection.** In our first SMTP encoding, we modelled the server's `OK`/`ERR` responses as branch (the server "offers" OK or ERR) rather than selection (the server internally "selects" the response). The state space was the same size, but the polarity was wrong—this matters for subtyping (Step 7) and duality (Step 8). *Lesson:* polarity is a semantic, not syntactic, property.

3. **Forgetting the exit.** An early version of the Raft encoding had no `SHUTDOWN` transition—the protocol looped forever. The state space was a single non-trivial SCC with no bottom element, violating boundedness. The well-formedness check (Step 1) caught this immediately. *Lesson:* every protocol needs a termination path.

These mistakes were valuable: they stress-tested the formalism and helped us refine the encoding conventions of Section 2.2.

8.3 On Encoding Fidelity

A legitimate concern is whether our session-type encodings faithfully represent the protocols they claim to model. We address this in three ways: (i) each benchmark includes a plain-language *problem statement* and a *protocol description* that a domain expert can review; (ii) the Hasse diagram provides a visual sanity check (does the state space "look right" for this protocol?); and (iii) the cross-tool validation ensures that the encoding is unambiguous (both tools produce the same state space from the same type string).

That said, every formalisation involves abstraction. Our SMTP encoding omits pipelining; our Raft encoding omits log compaction; our enzyme model omits allosteric regulation. These omissions are deliberate: we model the *core protocol structure*, not every implementation detail. The appropriate level of detail is a judgement call that depends on the intended application.

8.4 Scaling the Catalogue

The catalogue currently contains 79 binary and 24 multiparty protocols. We envision future expansion along several axes:

- **More domains**: finance (FIX protocol, payment processing), healthcare (HL7 FHIR workflows), IoT (MQTT, CoAP), and industrial control (OPC UA state machines).

- **Larger protocols**: protocols with 50+ states that stress-test the tools and reveal scalability limits.

- **Community contributions**: publishing the catalogue as a public repository where researchers can submit new benchmarks.

9 Conclusion

Step 2 has transformed the abstract state-space construction of Step 1 into a concrete, validated, and richly annotated benchmark catalogue. The 79 binary and 24 multiparty protocols span eight domains—from SMTP to stellar nucleosynthesis—and every one of them forms a lattice under the SCC quotient ordering. This universal lattice property is the central empirical finding that drives the rest of the programme.

The catalogue also reveals quantitative patterns: parallel composition roughly doubles state-space size; SCC compression removes about 23% of states in recursive protocols; 80% of benchmarks have distributive lattices; and structurally identical lattices arise independently in unrelated domains.

Looking ahead, Step 3 will begin the theoretical work of explaining *why* these lattices arise, by proving that external choice (branching) produces meets. The benchmarks will serve as the test bed: every theoretical claim will be validated against the full catalogue before publication.

Availability. All 79 binary benchmarks are defined in
`reticulate/tests/benchmarks/protocols.py`
24 multiparty benchmarks in
`reticulate/tests/benchmarks/multiparty_protocols.py`
Full catalogue with Hasse diagrams at
`docs/benchmark-catalogue/benchmark-catalogue-v2.pdf`

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